

Required Capstone Checkpoint 4.1: Advanced Retrieval Implementation

Directions

Download this worksheet and save a copy to your device before you begin working. Use this worksheet to plan, organize, and complete your submission for Required Capstone Checkpoint 4.1. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Completion Checklist

- ☐ Identification of a meaningful retrieval failure or limitation supported by evaluation evidence
- ☐ Analysis of the likely cause of the retrieval failure
- ☐ Selection and justification of an advanced retrieval strategy
- ☐ Evidence of implementation of the updated retrieval architecture
- ☐ Evaluation results comparing the updated system against the baseline system
- ☐ Reflection on the effectiveness of the architectural changes and any remaining limitations

Step 1. Retrieval Failure Identification

Identify a meaningful retrieval limitation in your current retrieval system.

- Describe the failure or limitation.
- Explain how the limitation affects retrieval performance.
- Reference evaluation evidence, test results, queries, or outputs that demonstrate the issue.

Retrieval failure or limitation

The dominant failures in the baseline retrieval system are very poor recall and precision when:

- (1) the retriever must find specific details buried in very long documents
- (2) the retriever is tasked with a query involving multiple entities
- (3) the retriever is tasked with a complex query involving an unnamed entity

Evidence supporting the identified limitation

The following results from testing the baseline retrieval system using the test suite for 4.1 (Capstone\Checkpoint 4.1\mini-4.1-test-suite.json), which are reported in Capstone\Checkpoint 4.1\evaluation_results\evaluation_baseline_20260921_161030.json, illustrate these failures:

(1) G16 “Was Charles III born before his mother’s reign began?”

The following screenshots show that the system was unable to answer this question and that all recall and precision metrics are 0.

Attribute	Data
id	1 G16
question	1 Was Charles III born before his mother's reign began?
information_need	1 fact_lookup
motivation	1 conversation
prior_familiarity	1 familiar
scope	1 multi_document
answerable	1 true
adversarial_kind	1
required_sources	1 [2 "Charles_III.html" 3]
expected_sources	1 [2 "Charles_III.html", 3 "Elizabeth_II.html" 4]
retrieved_sources	1 [2 "Louis_XV.html", 3 "Louis_XIV.html", 4 "Edward_III_of_England.html" 5]

required_aspects	1 [2 "The answer is yes.", 3 "Both dates (born 14 November 1948; accession 6 February 1952) are given." 4]
response	1 The documents do not provide enough information to determine whether Charles III was born before his mother's reign began.
closed_book_response	1 No. Charles III was born on November 14, 1948, and his mother, Queen Elizabeth II, began her reign on February 6, 1952. So he was born before her reign began.
correctness	1 fail
graded_correctness	1 incorrect
graded_correctness_score	1 0
covered_aspects	1 []
aspect_coverage	1 0
answer_relevance	1 pass
claims	1 [2 { 3 "text": "The documents do not provide enough information to determine whether Charles III was born before his mother's reign began.", 4 "cited_source": null, 5 "grounded": false, 6 "citation_supported": null, 7 "attributable_to_retrieval": null 8 }



claim_count	1 1
graded_groundness_score	1 0
faithfulness	1 fail
citation_support_rate	1
chunk_attribution_score	1
source_hit_at_k	1 0
source_recall_at_k	1 0
source_precision_at_k	1 0
source_recall_strict	1 0
all_required_present	1 false
response_quote_count	1 0
quote_fidelity_rate	1
quote_attribution_accuracy	1
citation_count	1 0
citation_precision	1
unsupported_citation_rate	1
refusal_outcome	1 n/a

(2) Q003 “Compare the Academy Awards ceremonies held a decade apart: who hosted the 85th and 95th ceremonies, and which film won Best Picture at each?”

In this instance, the system was able to provide a partial answer despite once again scoring 0 on the recall and precision metrics.

Attribute	Data
id	1 Q003
question	1 Compare the Academy Awards ceremonies held a decade apart: who hosted the 85th and 95th ceremonies, and which film won Best Picture at each?
retriever	1 baseline
information_need	1 fact_lookup
motivation	1 work_or_school
prior_familiarity	1 unfamiliar
scope	1 multi_document
answerable	1 true
adversarial_kind	1
required_sources	1 [2 "85th_Academy_Awards.html", 3 "95th_Academy_Awards.html" 4]
expected_sources	1 [2 "85th_Academy_Awards.html", 3 "95th_Academy_Awards.html" 4]
retrieved_sources	1 [2 "Academy_Award_for_Best_Director.html", 3 "Academy_Award_for_Best_Picture.html", 4 "Academy_Award_for_Best_Actor.html" 5]
retrieved_units	1 [2 { 3 "unit_id": "Academy_Award_for_Best_Director.html", 4 "source_file": "Academy_Award_for_Best_Director.html", 5 "role": "primary", 6 "score": 0.9018 7 }, 8 { 9 "unit_id": "Academy_Award_for_Best_Picture.html", 10 "source_file": "Academy_Award_for_Best_Picture.html", 11 "role": "primary", 12 "score": 0.8544 13 } 14]
unit_count	1 3
context_chars	1 112767
evidence_recall	1 0
retrieval_trace	1 {}
retrieval_latency_seconds	1 0.3270667999750003
source_hit_at_k	1 0
source_recall_at_k	1 0

source_recall_at_3_articles	1 0
source_precision_at_k	1 0
chunk_precision_at_k	1 0
article_precision	1 0
unique_articles_retrieved	1 3
source_recall_strict	1 0
all_required_present	1 false
required_aspects	1 [2 "Seth MacFarlane hosted the 85th Academy Awards, where Argo won Best Picture.", 3 "Jimmy Kimmel hosted the 95th Academy Awards, where Everything Everywhere All at Once won Best Picture." 4]
response	1 The provided documents do not say who hosted the 85th or 95th Academy Awards ceremonies, so I can't answer that part from the sources given. 2 3 For Best Picture: 4 - 85th Academy Awards: "Argo" won Best Picture [Academy_Award_for_Best_Picture.html] 5 - 95th Academy Awards: "Everything Everywhere All at Once" won Best Picture [Academy_Award_for_Best_Picture.html]

(3) G13 “What was the former name of the university where Barack Obama earned his law degree?”

Here the system was able to determine that Barack Obama earned his degree at Harvard University but was unable to determine that the institution’s former name was Harvard College. Source recall at k and aspect coverage were both 0.5.

Attribute	Data
id	1 G13
question	1 What was the former name of the university where Barack Obama earned his law degree?
information_need	1 fact_lookup
motivation	1 intrinsic_learning
prior_familiarity	1 unfamiliar
scope	1 multi_document
answerable	1 true
adversarial_kind	1
required_sources	1 [2 "Barack_Obama.html", 3 "Harvard_University.html" 4]
expected_sources	1 [2 "Barack_Obama.html", 3 "Harvard_University.html", 4 "List_of_presidents_of_the_United_States_by_education.html" 5]

retrieved_sources	1 [2 "Barack_Obama.html", 3 "List_of_presidents_of_the_United_States_by_education.html", 4 "Oprah_Winfrey.html" 5]
required_aspects	1 [2 "Harvard is identified as where Obama earned his law degree.", 3 "Harvard College is given as the former name." 4]
response	1 Barack Obama earned his law degree from Harvard Law School, and the documents say he enrolled there in 1988 and graduated in 1991 [Barack_Obama.html]. The documents do not state a former name for that university, so I can't determine one from the provided material.
closed_book_response	1 Barack Obama earned his law degree at **Harvard Law School**. It was formerly known as the **Harvard Law School of Harvard University**--though in common usage, it has long been called **Harvard Law School**.
correctness	1 fail
graded_correctness	1 incorrect
graded_correctness_score	1 0
covered_aspects	1 [2 1 3]
aspect_coverage	1 0.5
answer_relevance	1 pass
claims	1 [2 { 3 "text": "Barack Obama earned his law degree from Harvard Law School.", 4 "cited_source": "Barack_Obama.html", 5 "grounded": true, 6 "citation_supported": true, 7 "attributable_to_retrieval": false 8 }, 9 { 10 "text": "The documents say he enrolled at Harvard Law School in 1988.", 11 "cited_source": "Barack_Obama.html", 12 "grounded": true,
claim_count	1 5
graded_groundness_score	1 0.6
faithfulness	1 fail
citation_support_rate	1 0.75
chunk_attribution_score	1 0
source_hit_at_k	1 1
source_recall_at_k	1 0.5
source_precision_at_k	1 0.6666666666666666
source_recall_strict	1 0
all_required_present	1 false

Step 2. Failure Analysis

Analyze why the retrieval failure occurs. Consider questions such as:

- Does the query require multiple retrieval steps?
- Is relevant information distributed across multiple documents, conversations, or entities?
- Is excessive irrelevant information being retrieved?
- Are important relationships within the data not represented effectively?
- Are there other factors contributing to the failure?

Likely cause(s) of the retrieval failure

(1) G16 “Was Charles III born before his mother’s reign began?”

In this case the retriever matched the keywords “III / born / mother / reign” matched multiple monarch pages but none of the pages needed. The page for Charles III is 149k long.

(2) Q003 “Compare the Academy Awards ceremonies held a decade apart: who hosted the 85th and 95th ceremonies, and which film won Best Picture at each?”

In this case the partial answer was enabled by retrieving the source

“Academy_Award_for_Best_Picture.html”, but it was unable to answer who hosted the two award ceremonies, since it failed to retrieve the required documents “85th_Academy_Awards.html” and “95th_Academy_Awards.html”. There are many documents in the corpus about the Academy Awards but with TOP_K=3 the chances of getting the exact documents needed to make the comparison is slim.

(3) G13 “What was the former name of the university where Barack Obama earned his law degree?”

In this case retrieval was partially successful, retrieving two of the three expected sources: “Barack_Obama.html” and “List_of_presidents_of_the_United_States_by_education.html”, but because the query did not include “Harvard”, the retriever failed to retrieve “Harvard_University.html”, instead picking up “Oprah_Winfrey.html”, which contained several references to Barack Obama, whom Oprah endorsed in 2008 for President.

Why the current retrieval architecture struggles

The current architecture struggles when it is forced to search multiple, sometimes very long, documents with significant overlapping vocabulary for very specific details, but is configured to surface only 3 documents (TOP_K=3), or when the entity connecting the most important documents is not actually named in the query. In all these cases both a lexical and a semantic search end up surfacing many potential candidate documents all competing for the top 3 slots. In the case of a query with an unnamed entity connecting two documents, it is happenstance whether an article about the unnamed entity itself will surface, because lexical and semantic searches are blind to the unnamed entity.

Step 3. Retrieval Strategy Selection

Select an advanced retrieval strategy designed to address the identified limitation. Examples may include:

- Query rewriting
- Query decomposition
- Retrieval fusion
- Multi-hop retrieval
- Relationship-aware retrieval
- Graph-based retrieval
- Another advanced retrieval approach

Selected retrieval strategy

Each of the three failures can be addressed using the following advanced retrieval strategies:

- (1) Text splitting and chunking
- (2) Query decomposition
- (3) Graph based retrieval

Justification for the selected strategy

- (1) Text splitting and chunking to create finer grained searchable units of text, retrieved using a higher TOP_K and CANDIDATE_POOL for hybrid search, so that the chunks most relevant to the query rise to the top
- (2) Decomposing a composite or comparison multi-entity query into sub-queries to increase the likelihood that the detail required about each entity can be retrieved and then fused to provide the required context

(3) Creation of a graph that establishes common relationships between entities to enable the system to go beyond the first hop to find what it needs

Step 4. Updated Retrieval Architecture

Describe the modifications made to your retrieval system.

- Summarize the architectural changes.
- Explain how the updated approach differs from the baseline system.
- Include implementation evidence where appropriate.

Description of architectural changes

I made two major changes: one to the Checkpoint 2.1 extraction module, one to the evaluation script carried over from Checkpoint 3.1:

(1) amended the HTML extraction with a new function (`process_wikipedia_with_nodes_and_edges`) to add to the basic text extraction per document (`extract_wikipedia_text`). The new function does text splitting and chunking using LangChain's `MarkdownHeaderTextSplitter` (to capture headings) and `RecursiveCharacterTextSplitter` (for long text), and uses Beautiful Soup to extract some of the semantic properties such as Wikilinks, categories, and infoboxes, to generate graph data (nodes and edges) for later use in building a network graph.

(2) amended the script to be run with one of four different retrievers (baseline, chunks, decompose, graph) using the `--retriever` argument. The baseline retriever is just the same code as the 3.1 checkpoint. The other retrievers use `process_wikipedia_with_nodes_and_edges`. If the script is called with "chunks" then a chunk index and chunk vector store is created and used for retrieval, which adopts a parent-child retriever model configured in these runs with `--parent lead_section`, which extracts the chunk that got the hit together with the rest of its section, plus the article's lead chunks and infobox. If the script is called with "decompose" then it builds upon the "chunks" option and decomposes the query into sub-queries and fuses the results. If the script is called with "graph" then a network graph is created from the `WikipediaGraphData` nodes and edges and the graph is used to bring in chunks from category edges, wikilinks edges, or infobox edges to add to the context.

Implementation evidence (e.g., screenshots, code snippets, retrieval outputs, notebook excerpts)

(1) Here are some screenshots of the updated extraction script, which can be found at Capstone\Checkpoint 2.1\Extract_text_and_links_from_wikis.py. For documentation see Capstone\docs\Extract_text_and_links_from_wikis.md.

```

49  from bs4 import BeautifulSoup
50  import trafilatura
51  from langchain_core.documents import Document
52  from langchain_text_splitters import MarkdownHeaderTextSplitter, RecursiveCharacterTextSplitter
53
54
55  @dataclass
56  class WikipediaGraphData:
57      """Graph-ready Wikipedia content without a dependency on a graph library."""
58
59      documents: list[Document]
60      nodes: list[dict[str, object]]
61      edges: list[dict[str, object]]
62
63
115  def extract_categories(soup: BeautifulSoup) -> list[tuple[str, str]]:
116      """Return (display name, link slug) for each real topical category on the
117      page, skipping hidden/maintenance categories (e.g. "Articles with short
118      description", "Use dmy dates from...") which carry no topical meaning.
119      """
120      links = soup.select(".mw-normal-catlinks a[href^='/wiki/Category:']")
121      results = []
122      for anchor in links:
123          slug = wiki_link_slug(anchor.get("href", ""))
124          if slug is None:
125              continue
126          results.append((anchor.get_text(strip=True), slug))
127      return results
128
219  def process_wikipedia_with_nodes_and_edges(
220      directory_path: str, max_chunk_chars: int = 2000
221  ) -> WikipediaGraphData:
222      """Extract document chunks plus normalized article nodes and hyperlink edges."""
223      all_chunks = []
224      nodes = []
225      edges = []
226

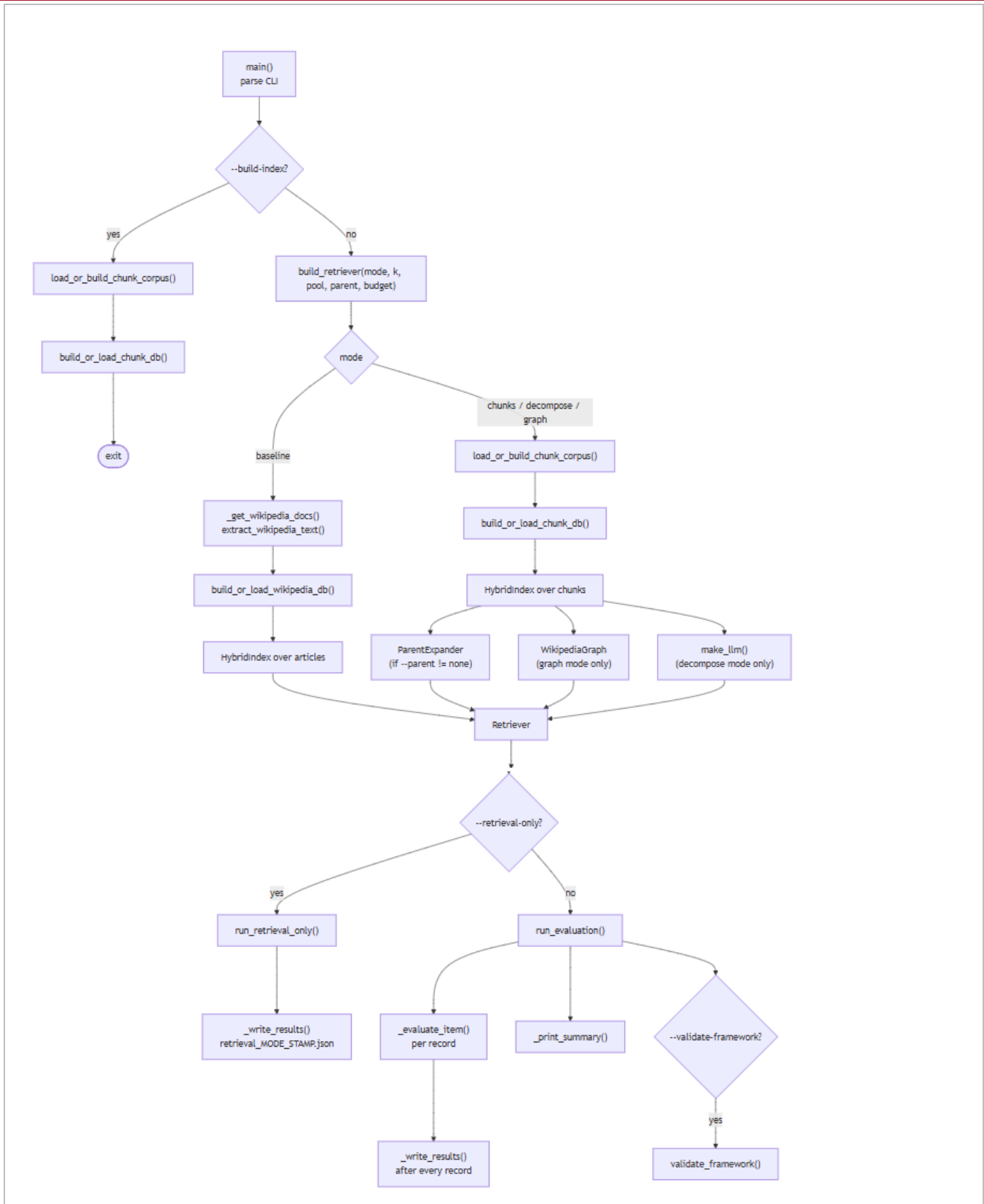
```

```

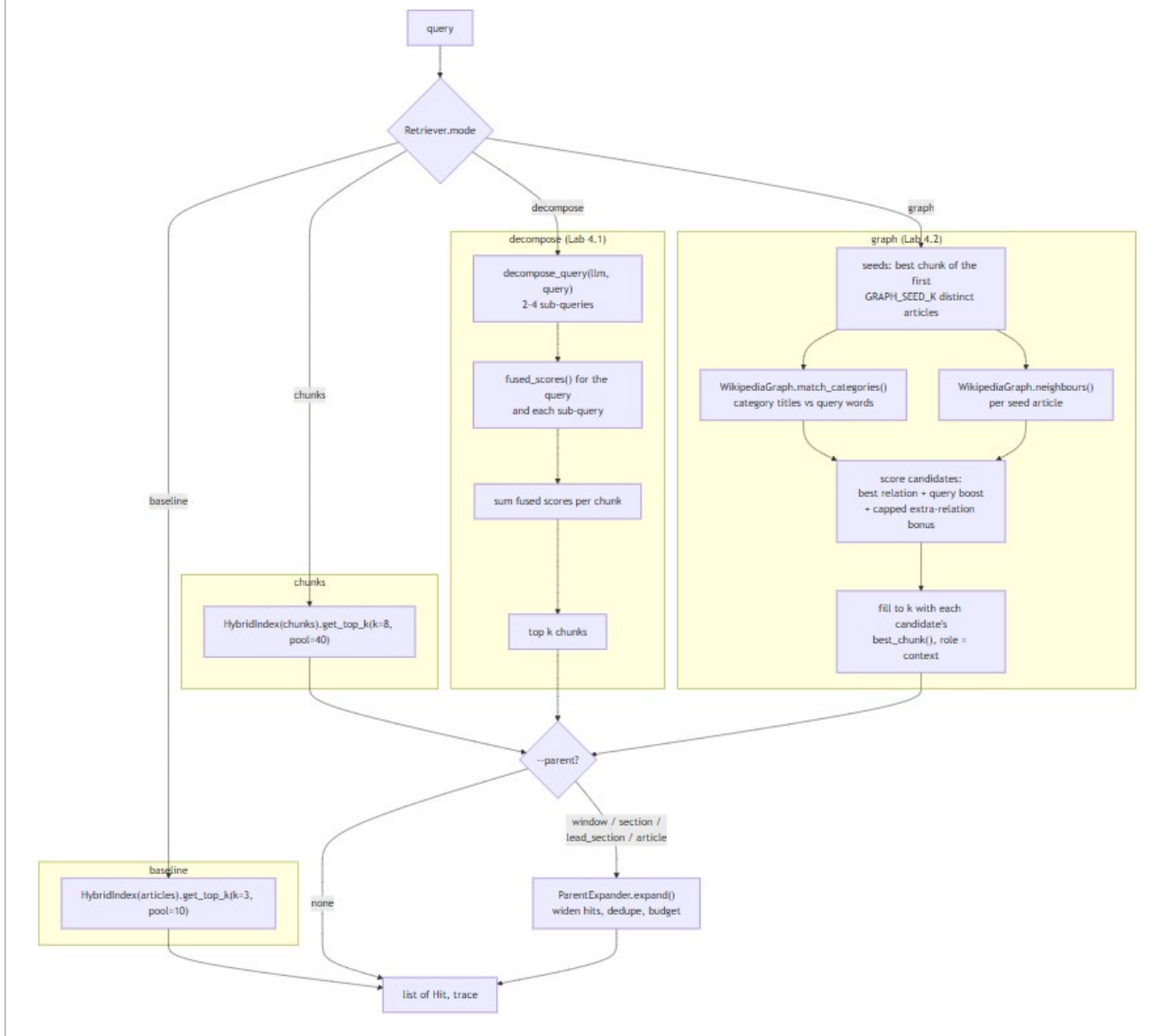
246 # 1. Split on Wikipedia's section headings. strip_headers=False keeps the
247 # heading text inside the chunk so BM25 and embeddings can see it.
248 headers_to_split_on = [("#", "Header 1"), ("##", "Header 2"), ("###", "Header 3")]
249 header_splitter = MarkdownHeaderTextSplitter(
250     headers_to_split_on=headers_to_split_on, strip_headers=False
251 )
252
253 # 2. Sub-split long sections.
254 text_splitter = RecursiveCharacterTextSplitter(
255     chunk_size=max_chunk_chars,
256     chunk_overlap=200,
257     separators=["\n\n", "\n", " ", ""],
258 )
259
260
261 # --- Title ---
262 soup = BeautifulSoup(html_content, "lxml")
263 page_title = soup.title.get_text(strip=True) if soup.title else None
264 if page_title:
265     page_title = page_title.removesuffix(" - Wikipedia")
266 article_title = page_title or filename.removesuffix(".html").replace("_", " ")
267
268 # --- Categories (article -> category edges) ---
269 categories = extract_categories(soup)
270 category_edges = []
271 for name, slug in categories:
272     category_id = category_ids.setdefault(
273         slug, hashlib.sha256(slug.encode("utf-8")).hexdigest()[:16]
274     )
275     category_edges.append({
276         "source": article_id,
277         "target": category_id,
278         "edge_type": "in_category",
279         "source_file": filename,
280         "category_name": name,
281     })

```

(2) Here is a flowchart of the main logic in the advanced retrieval system.



And here is a flowchart of the main query logic.



Step 5. Evaluation Results

Evaluate the updated system using the evaluation framework developed in previous modules.

- Present evaluation results.
- Compare performance against the baseline system.
- Include representative examples.

Evaluation Metric	Baseline	Chunks	Decompose	Graph
Binary correctness	7/17 (41%)	6/17 (35%)	7/17 (41%)	8/17 (47%)
Mean graded correctness	0.47	0.43	0.56	0.66
Mean aspect coverage	0.55	0.56	0.69	0.67
Faithfulness (all claims grounded)	10/17	10/17	11/17	9/17
Mean graded groundedness	0.79	0.87	0.85	0.86
Mean citation precision	0.66	0.76	0.85	0.78
Mean quote fidelity	0.67	1.00	0.92	0.79
Mean chunk attribution	0.60	0.62	0.48	0.59
Strict source recall (all required articles)	6%	41%	47%	59%
Source recall@3 articles (matched count)	24%	39%	55%	39%
Mean unique articles in context	3.0	5.1	4.2	8.0
Precision	37% (@3 articles)	48% chunk / 42% article	55% / 49%	26% / 26%
Mean evidence recall (golden quotes in context)	36%	50%	51%	50%
Mean context per query	225k chars	29k	25k	39k
Mean tokens per record	232k	33k	29k	44k
Cost (17 records)	\$3.00	\$0.46	\$0.40	\$0.60
Mean / p95 latency (all runs paced at ≥3.1 s between calls)	10.0 / 13.3 s	10.9 / 14.5 s	14.2 / 19.7 s	10.1 / 14.5 s

Compared to the baseline, every advanced retriever fared much better in terms of recall, but only the graph retriever fared better at binary correctness, even though its precision dropped significantly, while the decompose retriever fared better at graded correctness (0.47 to 0.56), and the chunk retriever actually fared worse on binary correctness. The decompose retriever fared best in precision and evidence recall (golden quotes) and citation precision. After chunking, costs dropped significantly for all three advanced retrievers.

Example 1: Query, retrieved information, and result

(1) G16 “Was Charles III born before his mother’s reign began?”

The following screenshots taken from Capstone\Checkpoint

4.1\evaluation_results\evaluation_chunks_20260921_164639.json show that with “chunks” recall flipped from 0 to 1 while precision went from 0.0 to 0.5. The retrieved context this time contained the birth date and the accession year. Despite this, the system produced an incorrect answer with “chunks” (a separate issue), but a correct answer with “decompose” and “graph”. The main takeaway is that the system found the right chunks to answer the question.

required_sources	<pre> 1 [2 "Charles_III.html" 3] </pre>
expected_sources	<pre> 1 [2 "Charles_III.html", 3 "Elizabeth_II.html" 4] </pre>
retrieved_sources	<pre> 1 [2 "Charles_III.html", 3 "James_II_of_England.html", 4 "List_of_Scottish_monarchs.html", 5 "Anne,_Princess_Royal.html", 6 "Nicola_Sturgeon.html" 7] </pre>
retrieved_units	<pre> 1 [2 { 3 "unit_id": "Charles_III.html#c00000-c00002", 4 "source_file": "Charles_III.html", 5 "role": "primary", 6 "score": 0.8501 7 }, 8 { 9 "unit_id": "James_II_of_England.html#c00000-c00003", 10 "source_file": "James_II_of_England.html". </pre>

unit_count	1 8
context_chars	1 25941
evidence_recall	1 0
retrieval_trace	<pre> 1 { 2 "child_chunk_ids": [3 "dc85b7cd3cc95787-c00000", 4 "cb0448e3c9ba4282-c00000", 5 "b2f40881843f6170-c00025", 6 "dc85b7cd3cc95787-c00066", 7 "dc85b7cd3cc95787-c00178", 8 "6cfe65423b4b68e0-c00002", 9 "6673399075a4a01c-c00073", 10 "dc85b7cd3cc95787-c00187" 11] </pre>
retrieval_latency_seconds	1 0.8607216999516822
source_hit_at_k	1 1
source_recall_at_k	1 1
source_recall_at_3_articles	1 1
source_precision_at_k	1 0.5
chunk_precision_at_k	1 0.5
article_precision	1 0.2

unique_articles_retrieved	1 5
source_recall_strict	1 1
all_required_present	1 true
required_aspects	1 [2 "The answer is yes.", 3 "Both dates (born 14 November 1948; accession 6 February 1952) are given." 4]
response	1 No. Charles III was born on 14 November 1948, and the document says he "was born during the reign of his maternal grandfather, King George VI" and "became heir apparent when his mother, Queen Elizabeth II, acceded to the throne in 1952" [Charles_III.html].
closed_book_response	1 No. Charles III was born on November 14, 1948, and his mother, Queen Elizabeth II, began her reign on February 6, 1952. So he was born before her reign started.
correctness	1 fail
graded_correctness	1 incorrect
graded_correctness_score	1 0
covered_aspects	1 [2 2 3]
aspect_coverage	1 0.5
answer_relevance	1 pass

Example 2: Query, retrieved information, and result

(2) Q003 "Compare the Academy Awards ceremonies held a decade apart: who hosted the 85th and 95th ceremonies, and which film won Best Picture at each?"

In this instance, as reported in Capstone\Checkpoint 4.1\evaluation_results\evaluation_decompose_20260921_181908.json and shown below, with "decompose" the system was able to provide a complete and accurately cited answer, flipping recall from 0 to 1 and improving precision from 0 to 0.4. The key takeaway is that decomposition enabled the system to retrieve relevant chunks from the required documents related specifically to the 85th and 95th award ceremonies.

required_sources	<pre> 1 [2 "85th_Academy_Awards.html", 3 "95th_Academy_Awards.html" 4] </pre>
expected_sources	<pre> 1 [2 "85th_Academy_Awards.html", 3 "95th_Academy_Awards.html" 4] </pre>
retrieved_sources	<pre> 1 [2 "95th_Academy_Awards.html", 3 "Academy_Award_for_Best_Picture.html", 4 "85th_Academy_Awards.html", 5 "List_of_awards_and_nominations_received_by_Steven_Spielberg.html", 6 "68th_Academy_Awards.html", 7 "74th_Academy_Awards.html" 8] </pre>
retrieved_units	<pre> 6 "score": 2.7798 7 }, 8 { 9 "unit_id": "7584ab2c6b497c73-c00000", 10 "source_file": "Academy_Award_for_Best_Picture.html", 11 "role": "primary", 12 "score": 2.0362 13 }, 14 { 15 "unit_id": "85th_Academy_Awards.html#c00000-c00029", 16 "source_file": "85th_Academy_Awards.html", 17 "role": "primary", </pre>

unit_count	1 7
context_chars	1 23150
evidence_recall	1 1
retrieval_trace	<pre> 6 "95th Academy Awards ceremony host" 7], 8 "child_chunk_ids": [9 "110a1f26b577e6e7-c00000", 10 "7584ab2c6b497c73-c00000", 11 "f03136cff601fca1-c00028", 12 "f03136cff601fca1-c00012", 13 "f03136cff601fca1-c00000", 14 "198df6a3abe7cc47-c00020", 15 "3e8e548f009bb2b4-c00001", 16 "54cc717f4dc58b8c-c00001" 17], </pre>
retrieval_latency_seconds	1 5.152943600027356
source_hit_at_k	1 1
source_recall_at_k	1 1
source_recall_at_3_articles	1 1
source_precision_at_k	1 0.42857142857142855
chunk_precision_at_k	1 0.42857142857142855
article_precision	1 0.3333333333333333

unique_articles_retrieved	1 6
source_recall_strict	1 1
all_required_present	1 true
required_aspects	1 [2 "Seth MacFarlane hosted the 85th Academy Awards, where Argo won Best Picture.", 3 "Jimmy Kimmel hosted the 95th Academy Awards, where Everything Everywhere All at Once won Best Picture." 4]
response	1 The 85th Academy Awards were hosted by Seth MacFarlane, and *Argo* won Best Picture [85th_Academy_Awards.html]. The 95th Academy Awards were hosted by Jimmy Kimmel, and *Everything Everywhere All at Once* won Best Picture [95th_Academy_Awards.html].
closed_book_response	1 The **85th Academy Awards** were hosted by **Seth MacFarlane**, and **Argo** won **Best Picture**. 2 3 The **95th Academy Awards** were hosted by **Jimmy Kimmel**, and **Everything Everywhere All at Once** won **Best Picture**.
correctness	1 pass
graded_correctness	1 complete
graded_correctness_score	1 1
covered_aspects	1 [2 1, 3 2 4]

Example 3 (optional)

(3) G13 “What was the former name of the university where Barack Obama earned his law degree?”

Here the system with the “graph” retriever was both able to determine that Barack Obama earned his degree at Harvard University and was able to determine that the institution’s former name was Harvard College. As the screenshots below from Capstone\Checkpoint 4.1\evaluation_results\evaluation_graph_20260921_183400.json show, both source recall at k and aspect coverage went up from 0.5 to 1 and the answer flipped from fail to pass.

required_sources	<pre> 1 [2 "Barack_Obama.html", 3 "Harvard_University.html" 4] </pre>
expected_sources	<pre> 1 [2 "Barack_Obama.html", 3 "Harvard_University.html", 4 "List_of_presidents_of_the_United_States_by_education.html" 5] </pre>
retrieved_sources	<pre> 1 [2 "Barack_Obama.html", 3 "Ben_Cardin.html", 4 "Ivy_League.html", 5 "List_of_presidents_of_the_United_States_by_education.html", 6 "George_W._Bush.html", 7 "Donald_Trump.html", 8 "Columbia_University.html", 9 "Harvard_University.html" 10] </pre>
retrieved_units	<pre> 1 [2 { 3 "unit_id": "Barack_Obama.html#c00000-c00003", 4 "source_file": "Barack_Obama.html", 5 "role": "primary", 6 "score": 0.9075 7 }, 8 { 9 "unit_id": "Ben_Cardin.html#c00000-c00004", </pre>

unit_count	1 8
context_chars	1 42132
evidence_recall	1 0.6666666666666666
retrieval_trace	<pre> 85 { 86 "source_file": "Harvard_University.html", 87 "relations": [88 "'Education' of Barack_Obama.html", 89 "linked from Ivy_League.html", 90 "Ivy_League.html is 'Sporting affiliations' of Harvard_University.html", 91 "linked from List_of_presidents_of_the_United_States_by_education.html" 92], 93 "score": 6.827 94 } 95], 96 "child_chunk_ids": [</pre>
retrieval_latency_seconds	1 2.006987800067547
source_hit_at_k	1 1
source_recall_at_k	1 1
source_recall_at_3_articles	1 0.5
source_precision_at_k	1 0.375
chunk_precision_at_k	1 0.375
article_precision	1 0.375
unique_articles_retrieved	1 8
source_recall_strict	1 1
all_required_present	1 true
required_aspects	<pre> 1 [2 "Harvard is identified as where Obama earned his law degree.", 3 "Harvard College is given as the former name." 4] </pre>
response	<pre> 1 Barack Obama earned his law degree at Harvard Law School, which is part of Harvard University [Barack_Obama.html][Harvard_University.html]. The provided documents do not state a former name for Harvard University itself, only that its former name was "Harvard College" [Harvard_University.html]. </pre>
closed_book_response	<pre> 1 Barack Obama earned his law degree from **Harvard Law School**. Its former name was the **Harvard Law School of Harvard University** commonly just referred to as **Harvard Law School**; it has not had a widely used different former institutional name in modern times. 2 3 If you meant the university he attended before law school, that was **Columbia University**. </pre>
correctness	1 pass
graded_correctness	1 complete
graded_correctness_score	1 1
covered_aspects	<pre> 1 [2 1, 3 2 4] </pre>

Step 6. Architectural Impact Analysis

Analyze the impact of the retrieval strategy.

- What improved?
- What remained challenging?
- Did the strategy address the original failure?

Improvements observed

All three advanced retrieval architectures showed significant improvements in recall (strict recall (all required articles) rose from 6% to 41%/47%/59%, and in the case of chunking and decomposition also with precision (which rose from 37% to 48% and 55% respectively). Correctness improved in the graph case from 7 to 8 but fell slightly with chunking alone from 7 to 6. The cost per record dropped from 18 cents to between 2 and 4 cents.

Remaining limitations or challenges

In this checkpoint I focused exclusively on retrieval. This still needs tuning, since each technique also had its losses: the chunker's parent policy failed to retrieve the right chunk in the case that flipped from pass to fail, decomposition lost in two questions on single aspects, and precision dropped significantly for several questions using the graph because of taking in too many irrelevant adjoining nodes (graph expansion needs to be constrained). Dynamically expanding parent scope might fix parent policy problems.

But there are still several problems with the generation side, including the decomposition step naming answer entities from the model's own knowledge when writing sub-queries, which chunk attribution detects (0.62 → 0.48). Some of these generation problems could likely be fixed by tightening prompts (e.g. the decomposition prompt doesn't instruct against using prior knowledge – the answer prompt does) and reducing temperature from 0.2 to 0.0.

Did the strategy address the original failure?

The strategy addressed some of the original failures but in a piecemeal fashion. On the whole chunking brought a significant improvement to retrieval across the board, as well as slashing cost.

Step 7. Reflection and Next Steps

Reflect on the effectiveness of the architectural redesign.

- What did you learn from implementing this strategy?
- What would you improve next?
- Are there additional retrieval failures that still need to be addressed?

Reflection on the architectural redesign

Checkpoint 3.1's insights plus the lecture for module 4 led directly to a new test suite tailored to highlighting the gaps in the base retriever. This involved a deeper examination of the structure of Wikipedia articles and their semantic properties, the need to break the articles down into chunks to facilitate searches, and the introduction of new retrieval strategies to better handle multi-document and relational queries. Many of the required building blocks are now in place in the system.

Specific improvements or next steps

Each advanced retriever is adept at fixing specific problems related to a specific type of query but also introduces its own problems. To integrate these retrievers, what is needed is an orchestrator or router which analyzes the query and chooses the best retriever for the job, with a conditional edge which allows for further retrieval when required. And each retriever needs to be fine-tuned to handle regressions. Beyond that, several generation problems still need to be addressed with better prompting and tuning of hyperparameters. And interactive sessions will require a different test suite that accounts for the chat history and how this affects the LLM's context.

End of Required Capstone Checkpoint 4.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.